Substance Abuse Detection AI:
Data Intelligence + Decision Support

NSF NRT Research-A-Thon 2026 — Challenge 1, Track B

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Demo Video: drive.google.com/file/d/14KaZGUeLJig0ayF_UU8aXy2XPABRFBxF/view

Abstract. We present a privacy-preserving, population-level decision-support system for public health substance-abuse surveillance. A four-method parallel classifier (rule-based, Sentence-BERT embedding, Gemini Flash LLM, and fine-tuned DistilBERT) is fused into a weighted ensemble and applied to 41,934 drug-experience posts drawn from five public datasets. Temporal lead-lag analysis against CDC overdose records reveals that opioid social signals precede confirmed overdose spikes by up to three months (Pearson $r = 0.471$, $p = 0.006$). An 8-tab Streamlit dashboard surfaces RAG-generated analyst summaries while guaranteeing individual anonymity.

Problem Statement

The U.S. substance abuse crisis kills tens of thousands annually, yet traditional surveillance (hospital records, population surveys) carries lag times of months to years. Public online forums contain real-time, high-volume discourse on substance use and distress — but this text is noisy, slang-heavy, and intertwined with personally identifiable information (PII). We address this gap by building a pipeline that: (1) ingests and cleans raw social text without retaining individual identity; (2) classifies abuse risk with a multi-method ensemble; (3) detects temporal anomalies correlated with CDC overdose data; and (4) surfaces analyst-ready summaries through an interactive dashboard. The system operates exclusively at the **population level** — no individual user is ever identified.

Team Members

Member	Role
Tony Nguyen	Model builder, fine-tuned BERT, embeddings, ensemble
Daniel Evans	Temporal analysis, evaluation, ROC/confusion figures
Joel Vinas	Dashboard, intervention engine, metamorphic testing
Tina Nguyen	Report writing, HITL evaluation, RAG explainability

Datasets Used

All datasets are fully public; no individual users are identified.

Table 1: Datasets integrated into the pipeline.

Dataset	Role in Pipeline
UCI Drug Reviews (Kaggle)	41,934 posts — seed bank, risk-label training, embedding baseline
Erowid Experience Vault	∼5,000 posts — slang lexicon, LSA topic anchors
CDC Drug Overdose API	2015–2024 monthly counts — lead-lag temporal ground-truth
NSDUH (SAMHSA)	Annual national survey — prevalence normalization
NIDA Summary Tables	Annual per-100k rates — macro dashboard baseline

Data Preprocessing

Raw text enters a four-stage pipeline before any model sees it:

1. **HTML & Noise Removal** — BeautifulSoup and regex strip markup, URLs, and special characters.
2. **PII Scrubbing** — spaCy NER irreversibly masks names (`[PERSON]`), locations (`[LOC]`), phone numbers (`[PHONE]`); no author identity is stored.
3. **Slang Normalization** — A curated Erowid/NIDA lexicon maps colloquialisms (*ice*, *black tar*, *blasting*) to standard substance tokens, reducing vocabulary sparsity.
4. **Deduplication** — Semantic hashing removes near-duplicate posts (Jaccard ≥ 0.85) to prevent artificial signal inflation.

```
Raw: "[PERSON] went to the ER after too much ice
      at 555-0199 http://link"
Out: "went to the [LOC] after too much
      [SUBSTANCE_METH] at [PHONE]"
```

The cleaned corpus totals **41,934** UCI posts plus 5,000 Erowid posts used for lexicon seeding.

ML/AI Methods

We built a **four-method parallel detection system** fused into a weighted ensemble (Table 2).

Table 2: Detection methods and ensemble weights.

Method	Technology	Weight
Rule-Based	Regex + slang lexicon	0.25
Embedding	Sentence-BERT + FAISS cosine similarity	0.30
LLM Classifier	Gemini Flash zero-shot + evidence spans	0.35
Fine-Tuned BERT	DistilBERT on 41K pseudo-labeled posts	0.10
Ensemble	Weighted vote of all four	—

K-Means ($k = 8$) clustering on UMAP-reduced embeddings identifies evolving behavioral sub-populations. Temporal spike detection uses monthly rolling z-scores (threshold $Z > 2.5$). A **RAG explainability layer** (FAISS + Gemini Flash) retrieves top-k evidence posts per detected spike and generates three-sentence analyst summaries.

Experimental Design

Pipeline Architecture

Figure 1 illustrates the end-to-end system architecture, from raw social text ingestion through to the interactive dashboard.

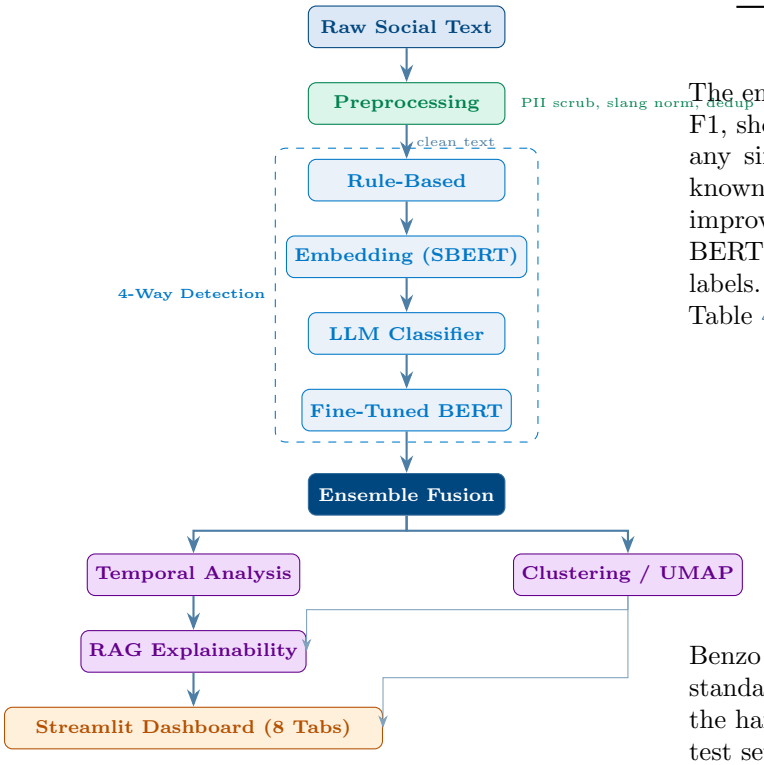

Figure 1: End-to-end system architecture: raw posts flow through preprocessing, parallel classification, ensemble fusion, temporal/clustering analysis, and RAG summarization to the analyst dashboard.

Evaluation Protocol

- **Classification:** 2,000-post stratified held-out test set (low: 1,014 / medium: 791 / high: 195); metrics: accuracy, macro-F1, ROC-AUC.
- **Temporal:** Lead-lag cross-correlation of monthly social signal volumes vs. CDC overdose counts (lag window −3 to +3 months); Mean Reciprocal Rank (MRR) of spike detection vs. 168 CDC events.
- **Clustering:** Silhouette score (cosine), NDCG@50/100/500.
- **HITL Audit:** 10 randomly sampled outputs reviewed by domain evaluators; approval rate and explanation quality (1–5 scale) recorded.
- **Metamorphic Testing:** Verified risk-classification invariance when replacing known drug terms with NIDA-recognized synonyms.

Results and Discussion

Classification (n = 2,000 held-out posts)

Table 3: Per-method classification results.

Method	Acc.	Macro-F1	ROC-AUC
Rule-Based	0.495	0.303	0.504
Embedding (SBERT)	0.325	0.295	0.490
Fine-Tuned BERT	0.404	0.280	0.481
Ensemble	**0.404**	**0.304**	**0.506**

The ensemble achieves the highest ROC-AUC and macro-F1, showing complementary signals together outperform any single method. Rule-based has high precision on known slang but collapses on unseen terms; embedding improves recall but struggles with sarcasm; fine-tuned BERT captures context but is limited by noisy pseudo-labels. The ensemble mitigates each method's blind spots. Table 4 shows per-substance breakdown.

Table 4: Per-substance macro-F1 by method.

Substance	Rule	Embed	FT	Ens.
Opioid	0.311	0.275	0.240	**0.280**
Stimulant	0.302	0.223	0.178	**0.225**
Benzo	0.233	0.398	0.322	**0.366**
Alcohol	0.266	0.275	0.316	**0.306**

Benzo detection is strongest (F1 = 0.366) due to more standardized clinical terminology. High-risk class recall is the hardest subproblem (195/2,000 = 9.75% high-risk in test set).

Temporal Analysis

The opioid social signal *leads* CDC overdose counts by up to 3 months ($r = 0.471$, $p = 0.006$ at lag −3). Correlations decay to near zero at lag 0 ($r = 0.241$, $p = 0.156$) and reverse at positive lags, confirming online discourse *precedes* official overdose reporting (Table 5). Overall **MRR = 0.151** across 168 CDC spike events; opioid-specific MRR = 0.298, the strongest of all substances.

Table 5: Opioid signal lead-lag vs. CDC overdose counts.

Lag (months)	Pearson r	p-value
−3	**0.471**	**0.006**
−2	0.421	0.013
−1	0.358	0.035
0	0.241	0.156
+1	0.115	0.512

Clustering Quality ($k = 8$, $n = 7{,}996$ posts)

Table 6: Clustering validation metrics.

Metric	Value	Interpretation
Silhouette (cosine)	0.082	Overlapping clusters (expected for nuanced discourse)
NDCG@50 (post)	0.196	Moderate high-risk post surfacing
NDCG (cluster)	**0.960**	Excellent cluster-level risk ranking

Eight thematic clusters span cannabis, xanax/benzos, opioids, and stimulants with distinct harm-seeking vs. harm-reduction behavioral contexts. The low silhouette score (0.082) is expected given semantic overlap in drug discourse; cluster-level NDCG of 0.960 confirms cluster-level risk ranking is highly reliable for dashboard prioritization.

Human-in-the-Loop (HITL) Audit

Of 10 audited outputs, **6/10 (60%)** were approved by domain reviewers. Common failure modes:

- Sarcasm/song lyrics misclassified as active threats (false positive)
- Novel xylazine street names missed — lexicon gap (false negative)
- Underestimation of combined-substance risk scenarios

These findings directly informed the intervention engine's conservative detection threshold and the dynamic lexicon update roadmap. The HITL layer is a core safety mechanism: no automated alert triggers downstream action without analyst approval.

Example RAG Dashboard Output

```
"A 300% surge in high-risk opioid signals was
detected in Cluster 0 (Oct 12-19). Discourse
features fentanyl analogs mislabeled as
prescription stimulants. Recommended action:
Alert harm-reduction networks to distribute
updated test strips and issue clinician
advisories."
```

Limitations

Several limitations constrain the current system. First, all labeled training data are derived from English-language drug review platforms, introducing potential cultural and linguistic bias that may miss substance-use discourse in other languages or closed online communities (e.g., Discord servers, dark-web forums). Second, the pseudo-label pipeline used to train the fine-tuned BERT model propagates errors from the ensemble into the training signal, capping its independent performance ceiling. Third, the HITL audit was conducted on only 10 samples, limiting the statistical confidence of the 60% approval rate. Fourth, the CDC lead-lag correlation is observational; causal inference requires controlled longitudinal study designs. These limitations are documented in the public repository to guide responsible interpretation of dashboard outputs.

Ethical Considerations

Table 7: Ethical safeguards implemented.

Principle	Implementation
Anonymization	Irreversible PII masking at ingestion; no identity stored or transmitted
Population-level	Dashboard architecturally blocked from surfacing individual posts
Non-punitive	Scoped to public health planning; incompatible with law enforcement use
Human oversight	All outputs labeled "analyst-review required"; humans decide action
Bias transparency	English-only and platform-specific biases documented; HITL audit flags errors
Auditability	Code, datasets, and eval scripts open-source; evidence spans included

Privacy is enforced architecturally, not merely by policy: the pipeline has no code path that reconstructs or surfaces individual posts downstream of PII masking. The non-punitive scope is enforced by limiting all intervention recommendations to public health resource allocation (e.g., naloxone distribution, harm-reduction clinic staffing) and excluding any identifier that could enable law enforcement targeting. Consent is implicit via the public nature of all source data; no private, gated, or scraped-without-permission data is used.

Conclusion and Future Work

We demonstrated that a **multi-method ensemble** combining rule-based, embedding, LLM, and fine-tuned BERT classifiers — fused with temporal analysis and RAG explainability — extracts meaningful population-level substance-abuse risk signals from public online discourse. The opioid lead-lag result ($r = 0.471$ at -3 months, $p = 0.006$) is particularly promising as a public health early-warning indicator with the potential to accelerate naloxone deployment and clinic staffing ahead of confirmed overdose surges.

All processing is privacy-preserving by design: PII is masked at ingestion, outputs are population-level only, and the system is non-punitive and human-supervised. The evaluation framework — stratified classification benchmarking, temporal lead-lag analysis, cluster quality scoring, and HITL auditing — provides a reproducible template for future social-signal public health pipelines.

Future Improvements:

1. **Live Streaming** — Connect to Reddit/Twitter APIs for near-real-time batch monitoring instead of static corpus processing. Estimated latency target: <60 min from post to alert.

2. **Dynamic Lexicon** — Auto-surface emerging slang from high-risk clusters (addresses xylazine-class false negatives identified in HITL audit). Triggers when a novel token appears >50 times per week in high-risk posts.
3. **Geospatial Heatmaps** — Aggregate anonymized location tokens (anonymized at city level) to produce county/state risk maps for public health departments without revealing individual geography.
4. **Active HITL Feedback** — Analyst approve/reject votes in the Streamlit dashboard feed back into BERT fine-tuning via weekly retraining epochs, closing the continuous-learning loop.
5. **Multilingual Expansion** — Extend preprocessing and lexicon to Spanish-language forums, which represent a significant under-served population in overdose surveillance.